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Exploring Useful Predictive Representations for Streaming RL under Partial Observability and Non-Stationarity

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PROBLEM STATEMENT

When does predictive knowledge become a useful state representation for replay-free streaming control?

We test a three-stage bottleneck: useful predictive state must preserve action-relevant distinctions, expose them to the learner, and remain compact and complementary under drift.

WHY THIS PROBLEM?

Accurate prediction ≠ useful control state

The Alberta Plan proposes predictions as agent state. In a partially observable Markov decision process (POMDP), retained hidden information may still be unusable by the online controller. We therefore separate three failure modes—preservation, accessibility, and adaptation under drift—rather than rank predictions alone.

PRESERVE		EXPOSE		ADAPT
retain the hidden action distinction	→	make it learnable by the online controller	→	keep a compact basis as the world drifts

RUNNING EXAMPLE

Continuing delayed-cue T-maze

cue L/R | ? | ? | ? | ? | ? | J — turn L / turn R

aliased corridor

Cue appears once. Corridor observations repeat. The junction requires the past cue.

RL task: cue → aliased corridor → junction turn; correct / incorrect feedback
Goal: continuing accuracy · Drift: cue corruption · distractors · timing · remapping

HOW WE TEST THE CLAIM

$$x_t = [\, o_t \,,\ f(g_t)\,,\ 1 \,]$$

g_t: learned forecasts from general value functions (GVFs)—cue, timing, regime
f: identity, residual random Fourier features (RFF), tile coding, or bank interface

RQ1 Representation geometry

4 POMDPs · 10 conditions · 3 seeds; compare transforms within environment. Accuracy and reward differ across tasks, so results are never pooled.

RQ2 Control accessibility

Trace-only T-maze · 10 paired seeds; hold g_t and the learner fixed, then vary only the controller adapter: identity, RFF, or tile coding.

RQ3 Bank structure under drift

Big-world T-maze · 4 seeds; vary bank k, diversity, and novelty, then compare against raw observations and a single cue-GVF baseline.

ONLINE GVF BANK UNDER DRIFT

A compact predictive division of labor, updated online

1 GENERATE		2 SCORE		3 RETAIN + EXPOSE
memory · timing control · regime + sparse mixtures	→	decision / decode gains + uncertainty reduction − TD error / redundancy	→	keep k complementary GVFs [o_t, bank, 1] → streaming linear policy

Mutate horizon / step size; retain candidates that fill missing roles and replenish online.

EXPERIMENTAL CONTROLS

- One-pass online learning; no replay, offline fitting, or recurrent memory.
- Held-out probes are diagnostic only; budgets and seeds are matched within comparisons.
- Environment-specific metrics; n=3/4 summaries are descriptive, not significance tests.

READOUTS **Decoding → preservation · post-switch → control · recovery → adaptation**

WHY THE SEQUENCE MATTERS

RQ1 tests whether feature reshaping generalizes across hidden structures. RQ2 fixes predictions to isolate controller accessibility. RQ3 asks how to compose predictions as relevance drifts. Together, the sequence moves from diagnosis to intervention to adaptive construction.

RESULTS

Three questions, three bounded answers

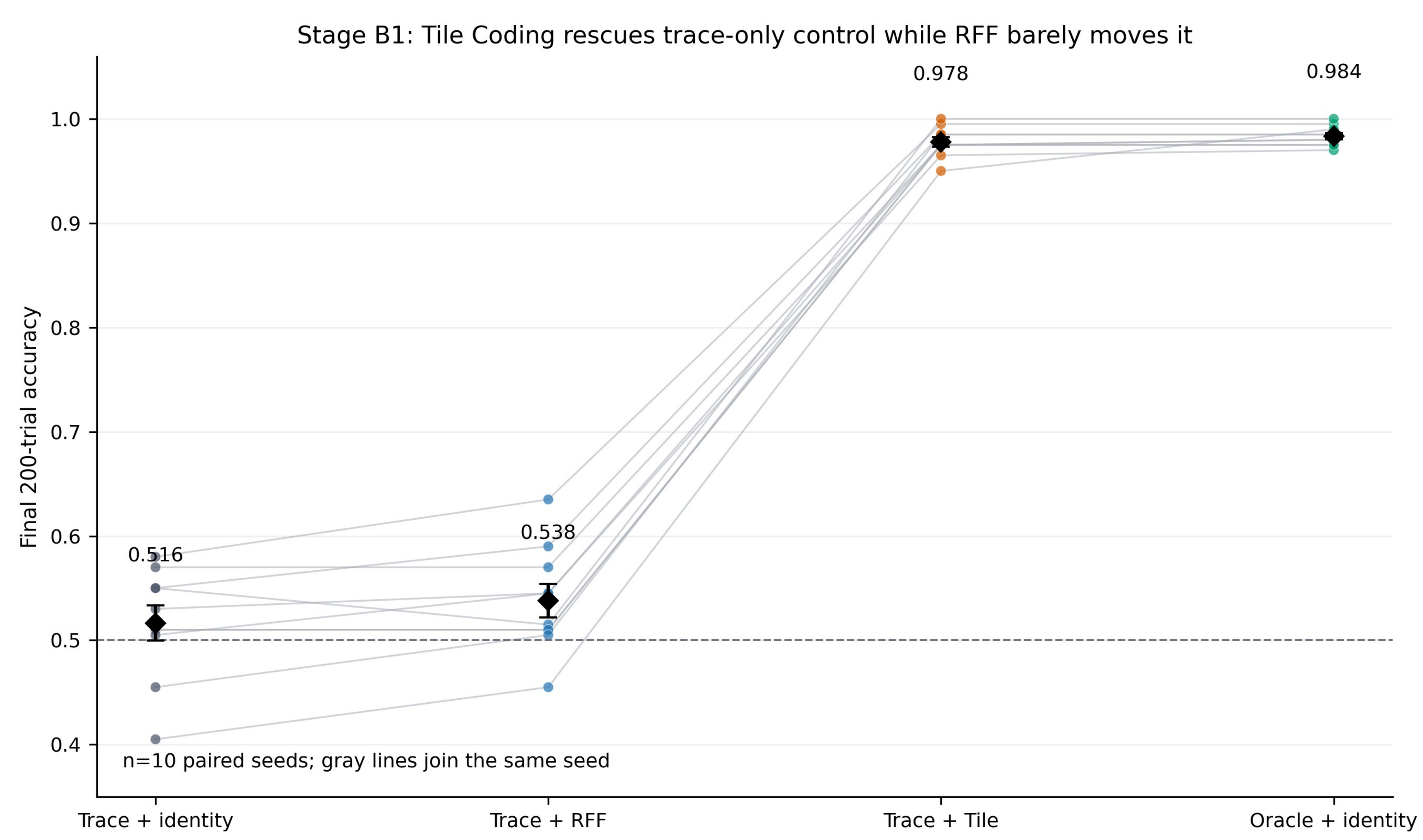
RQ1 · Does reshaping predictive features reliably improve control?

Environment	Selected result	What it rules out
T-maze	non-oracle ≈ 0.50 (chance)	feature shape alone does not recover memory control
Two-loop	**moment 0.797±0.068** **raw 0.495±0.004**	a positive effect can be environment-specific
Hidden velocity	raw −0.063±0.018 highest mean	regularization and priors can reduce control utility

Answer: no. Effects reverse across hidden structures; feature geometry is diagnostic, not a universal selection rule. With n=3/condition, the evidence supports counterexamples—not a general ranking.

RQ2 · Is decodable information accessible to online control?

The past cue is perfectly decodable at the junction (1.00); same g_t and learner, only f changes.



Identity 0.5165 · RFF 0.5380 · Tile 0.9780 · Oracle / direct cue 0.9835

Tile − identity = +0.4615 ± 0.0144 SEM · 10/10 paired improvements

Answer: not necessarily. Tile coding adds no cue information; local, sparse features expose the retained cue to the fixed linear learner, moving control near oracle—an accessibility, not prediction, gain.

RQ3 · What bank structure converts predictions into control under drift?

Raw 0.5073 · Cue GVF 0.5246 · four-seed means; uncertainty not supplied

Bank variant	Cue decodability	Post-switch control
k=2 full	0.6483	0.4950
k=3 full	**0.7294**	**0.5565**
k=4 full	0.7386	0.5408
k=3, no diversity	0.6951	0.5196
k=3, no novelty	**0.7294**	**0.5565**

- Remove diversity: −3.69 percentage points (0.5565 → 0.5196).
- k=4 decodes more (0.7386 > 0.7294) but controls worse (0.5408 < 0.5565).
- Remove novelty: no observed mean change (0.5565).

Answer: supplied means favor a compact, diversity-aware bank over increasing size or novelty alone. The k=4 counterexample shows that more decodable predictions can still add redundancy or interference.

CONCLUSION

In these streaming POMDPs, usefulness depends on the representation–learner interface—not on the predictive feature alone.

Same cue information, better access: tile coding moves control 0.5165 → 0.9780. More decodability, worse control: k=4 underperforms k=3—two counterexamples to selection by decoding alone.

Scope: controlled POMDPs · linear control · hand-structured GVF roles · n=3/4/10 by study. Next: jointly adapt bank composition and controller coupling.

REFERENCE

Sutton et al. · The Alberta Plan for AI Research · 2022

REPRODUCIBILITY

Streaming · replay-free · CPU-scale
Held-out probes are diagnostic only

SOURCE VERSIONS

Geometry 2940182f · Adapters a88813c
GVF bank: supplied 4-seed summaries
Commands/results accompany the course package